

| Distribution                | Type | Mass/density function $f(x)$                                                                                                                                          | Mean $\mu$                    | Variance $\sigma^2$                                         |
|-----------------------------|------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------|-------------------------------------------------------------|
| UNIFORM( $n$ )              | D    | $1/n$ , for $x = 1, 2, \dots, n$                                                                                                                                      | $(n+1)/2$                     | $(n^2-1)/12$                                                |
| UNIFORM( $a, b$ )           | C    | $\frac{1}{b-a}$ , for $x \in [a, b]$                                                                                                                                  | $\frac{a+b}{2}$               | $\frac{(b-a)^2}{12}$                                        |
| BERNOULLI( $p$ )            | D    | $f(0) = 1-p, f(1) = p$                                                                                                                                                | $p$                           | $p(1-p)$                                                    |
| BINOMIAL( $n, p$ )          | D    | $\binom{n}{x} p^x (1-p)^{n-x}$ ,<br>for $x = 0, 1, \dots, n$                                                                                                          | $np$                          | $npq$                                                       |
| GEOMETRIC( $p$ )            | D    | $q^{x-1}p$ , for $x = 1, 2, \dots$                                                                                                                                    | $1/p$                         | $(1-p)/p^2$                                                 |
| NEGATIVEBINOMIAL( $p, r$ )  | D    | $\binom{x-1}{r-1} p^r q^{x-r}$ ,<br>for $x = r, r+1, \dots$                                                                                                           | $r/p$                         | $r(1-p)/p^2$                                                |
| HYPERGEOMETRIC( $N, M, n$ ) | D    | $\frac{\binom{M}{x} \binom{N-M}{n-x}}{\binom{N}{n}}$ ,<br>for $x = 0, 1, \dots, n$                                                                                    | $np$                          | $np(1-p)$                                                   |
| POISSON( $\lambda t$ )      | D    | $\frac{1}{x!} (\lambda t)^x e^{-\lambda t}$ , for $x = 0, 1, \dots$                                                                                                   | $\lambda t$                   | $\lambda t$                                                 |
| GAMMA( $\lambda, r$ )       | C    | $\frac{1}{\Gamma(r)} \lambda^r x^{r-1} e^{-\lambda x}$                                                                                                                | $r/\lambda$                   | $r/\lambda^2$                                               |
| GAMMA( $\alpha, \beta$ )    |      | $= \frac{x^{\alpha-1} e^{-x/\beta}}{\beta^\alpha \Gamma(\alpha)}$ ,<br>for $x \in [0, \infty)$                                                                        | $= \alpha\beta$               | $= \alpha\beta^2$                                           |
| BETA( $\alpha, \beta$ )     | C    | $\frac{1}{B(\alpha, \beta)} x^{\alpha-1} (1-x)^{\beta-1}$ ,<br>for $0 \leq x \leq 1$                                                                                  | $\frac{\alpha}{\alpha+\beta}$ | $\frac{\alpha\beta}{(\alpha+\beta)^2(\alpha+\beta+1)}$      |
| NORMAL( $\mu, \sigma^2$ )   | C    | $\frac{1}{\sigma\sqrt{2\pi}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right)$ ,<br>for $x \in \mathbf{R}$                                                               | $\mu$                         | $\sigma^2$                                                  |
| CHISQUARED( $\nu$ )         | C    | $\frac{x^{\nu/2-1} e^{-x/2}}{2^{\nu/2} \Gamma(\nu/2)}$ , for $x \geq 0$                                                                                               | $\nu$                         | $2\nu$                                                      |
| T( $\nu$ )                  | C    | $\frac{\Gamma(\frac{\nu+1}{2})}{\sqrt{\pi\nu} \Gamma(\frac{\nu}{2}) (1+x^2/\nu)^{(\nu+1)/2}}$ ,<br>for $x \in \mathbf{R}$                                             | 0                             | $\nu/(\nu-2)$                                               |
| F( $\nu_1, \nu_2$ )         | C    | $\frac{1}{B(\frac{\nu_1}{2}, \frac{\nu_2}{2})} \left(\frac{\nu_1}{\nu_2}\right)^{\nu_1/2} x^{\nu_1/2-1} (1+\frac{\nu_1}{\nu_2}x)^{-(\nu_1+\nu_2)/2}$ ,<br>for $x > 0$ | $\frac{\nu_2}{\nu_2-2}$       | $\frac{2\nu_2^2(\nu_1+\nu_2-2)}{\nu_1(\nu_2-2)^2(\nu_2-4)}$ |